

CCNA - Module 4: Routing and Switching (M4-A1)

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Section A — Concept Application

S1. Static vs. Dynamic Routing

Operational Differences: Static routing requires manual configuration of paths by the network administrator. It consumes negligible CPU/RAM and generates no protocol traffic overhead. Dynamic routing uses protocols (e.g., OSPF, RIP) to automatically discover routes, build routing tables, and update topologies dynamically, which consumes system resources and bandwidth.

Justification: For a small startup with only two routers and a topology that rarely changes, static routing is ideal. It avoids unnecessary control-plane overhead, simplifies configuration, ensures deterministic traffic paths, and minimizes router resource usage.

S2. RIP vs. OSPF

Comparison: RIP (Routing Information Protocol) is a distance-vector protocol limited to a maximum hop count of 15, periodic full routing table updates (every 30 seconds), and slow convergence. OSPF (Open Shortest Path First) is a link-state protocol using Dijkstra's algorithm, offering fast convergence and scalable multi-area architecture.

Recommendation: OSPF is recommended for a growing network of 5+ routers. Its fast convergence helps prevent routing issues during network changes, and its hierarchical design scales as new branch offices are added.

S3. Switch MAC Address Table & Frame Forwarding

Learning & Forwarding: When Host A sends a frame to Host B, the switch examines the Source MAC address and records it along with the ingress port in its MAC address table. If the Destination MAC address is unknown, the switch floods the frame out of all ports except the ingress port. Once Host B responds, the switch records Host B's MAC address and ingress port. Subsequent frames are directly unicast forwarded.

S4. VLAN Segmentation & 802.1Q Trunking

VLAN Purpose: VLAN segmentation logically isolates broadcast domains on a shared physical switch, improving security, reducing broadcast traffic, and enforcing access control between departments such as Marketing and Engineering.

802.1Q Trunking: An 802.1Q trunk adds a VLAN tag to Ethernet frames as traffic crosses the trunk link. The VLAN ID enables network devices to distinguish traffic from multiple VLANs over a single physical link.

S5. Inter-VLAN Routing & Router-on-a-Stick

Requirement: Devices in different VLANs reside on separate logical IP subnets and cannot communicate at Layer 2. Inter-VLAN routing (Layer 3 forwarding) is required to route traffic between these subnets.

Advantages of Router-on-a-Stick: Logical sub-interfaces over a single trunk link eliminate the need for multiple physical router interfaces and additional cabling. This saves interface costs and switch ports while making efficient use of existing hardware.

S6. Wireless Security & MAC Filtering

Security & Filtering: MAC filtering restricts association based on hardware addresses. However, MAC addresses can be spoofed, making MAC filtering insufficient as a primary security layer.

Best Approach: Use WPA3-Enterprise (or WPA2/WPA3-Personal with a strong passphrase) for authentication and encryption, combined with guest-network controls or 802.1X/RADIUS where appropriate. This provides stronger protection than maintaining static MAC ACLs.

Section B — Practical Configuration Tasks

Task 1: Configure Management and Staff VLANs

Configuration:

```
Switch# configure terminal
Switch(config)# vlan 10
```

```

Switch(config-vlan)# name Management
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Staff
Switch(config-vlan)# exit
Switch(config)# interface FastEthernet 0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# exit
Switch(config)# interface FastEthernet 0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# exit
Switch# show vlan brief

```

Task 2: Configure an 802.1Q Trunk Link

Configuration:

```

Switch(config)# interface GigabitEthernet 0/1
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20
Switch(config-if)# exit
Switch# show interfaces trunk

```

Task 3: Configure Router-on-a-Stick for Inter-VLAN Routing

Configuration:

```

Router# configure terminal
Router(config)# interface GigabitEthernet 0/0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface GigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# exit
Router(config)# interface GigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# exit

```

Task 4: Capture and Analyze Inter-VLAN Traffic (Summary)

- Ingress:** Host A sends an Ethernet frame to the switch. The switch associates it with VLAN 10 and sends it across the trunk with the 802.1Q VLAN tag.
- Routing:** The router receives the frame, removes the Ethernet/802.1Q framing, examines the destination IP, decrements TTL, and routes the packet.
- Egress:** The router creates a new Ethernet frame for VLAN 20. The switch receives it on the trunk, removes the VLAN tag for the access port, and forwards it to Host B.

Section C — Mini Capstone Project Report

Problem Diagnosis

Router-on-a-Stick connectivity failure after a power outage.

Fault-Isolation Process (Layer 2 / Layer 3)

- Local Ping Test:** Pinged the local default gateway (192.168.10.1) from the Marketing host — successful.
- Inter-VLAN Ping Test:** Pinged the Engineering default gateway (192.168.20.1) from the Marketing host — failed.
- Verification Commands:** `show ip interface brief` on the router and `show interfaces trunk` on the switch.

Root Cause Identified

Following the power outage reboot, the main physical interface (GigabitEthernet 0/0) on the router was administratively down, or the startup configuration was missing the `no shutdown` command.

Resolution

Executed no shutdown on GigabitEthernet 0/0 and saved the configuration using copy running-config startup-config.

Section D — AI-Augmented Learning

1. Prompt Given to AI

Generate a complete Cisco IOS configuration script for a Cisco Switch and Router to set up Inter-VLAN routing (Router-on-a-Stick). Configure VLAN 10 (Marketing) and VLAN 20 (Engineering) on the switch with access ports, set up an 802.1Q trunk link to the router allowing VLANs 10 and 20, and configure sub-interfaces on the router for VLAN 10 and VLAN 20.

2. Configurations

The following shows the original configuration and the corrected configuration used for the lab.

AI Original Configuration Script (Defective)

```
enable
configure terminal
vlan 10
vlan 20
interface FastEthernet0/1
    switchport mode access
    switchport access vlan 10
interface FastEthernet0/2
    switchport mode access
    switchport access vlan 20
interface GigabitEthernet0/1
    switchport mode trunk
exit

enable
configure terminal
interface GigabitEthernet0/0.10
    encapsulation dot1Q 10
    ip address 192.168.10.1 255.255.255.0
interface GigabitEthernet0/0.20
    encapsulation dot1Q 20
    ip address 192.168.20.1 255.255.255.0
exit
```

Corrected Configuration Script

```
enable
configure terminal
vlan 10
    name Marketing
vlan 20
    name Engineering
interface FastEthernet0/1
    switchport mode access
    switchport access vlan 10
interface FastEthernet0/2
    switchport mode access
    switchport access vlan 20
interface GigabitEthernet0/1
    switchport mode trunk
    switchport trunk allowed vlan 10,20
exit

enable
configure terminal
interface GigabitEthernet0/0
    no shutdown
exit
interface GigabitEthernet0/0.10
    encapsulation dot1Q 10
    ip address 192.168.10.1 255.255.255.0
interface GigabitEthernet0/0.20
    encapsulation dot1Q 20
    ip address 192.168.20.1 255.255.255.0
exit
```

3. Debugging Note

The original script did not enable the parent physical interface GigabitEthernet 0/0 with no shut down, which can leave the sub-interfaces unable to pass traffic. The corrected configuration enabled the physical interface and explicitly restricted the trunk to VLANs 10 and 20. The changes restored the required link and VLAN connectivity.